

CS2204 Fundamentals of Internet Application Development
Course Work No. 3 (CW3)
Semester A 2025-2026

13% Marks

Due Date: 5 December 2025 (Friday), 23:59

Learning Outcomes:

- Able to write JavaScript and execute them on Web pages
- Able to enhance visual interest, access data, and manipulate elements of Web pages by applying JavaScript to appropriate HTML and CSS components.

1. Overview

Web pages prepared in CW1 and CW2 will be used. Enhance their interactivities and make all specified features work. JavaScript will be added, and you could also change the HTML/CSS in CW2, if necessary, to facilitate the functioning of JavaScript. Detail requirements are listed in the following sections.

To ensure timely feedback to the student's questions, each of the following TAs will be responsible for different groups of students according to **the session of the tutorial**. If you have any questions, you can contact the corresponding TA directly by email. Each TA will also attend the corresponding tutorials in person to answer your questions related to the assignment.

- **T01(Wen. 17:00-18:00): FAN Tianrui** (tianrufan2-c@my.cityu.edu.hk)
- **T02(Mon. 12:00-13:00): YUE Zhaoxi** (zhaoxiyue2-c@my.cityu.edu.hk)
- **T05(Mon. 13:00-14:00): LI Yuan** (yli2366-c@my.cityu.edu.hk)
- **T04(Mon. 14:00-15:00): LU Lu** (lulu62-c@my.cityu.edu.hk)
- **T03(Mon. 15:00-16:00): SUN Jiawen** (jiawensun7-c@my.cityu.edu.hk)

Guideline for late submissions:

- Late within **3 hours** - **10% marks deduction**
- Late within **3 ~ 12 hours** - **50% marks deduction**
- After **12 hours** - **no** submission will be accepted

2. External JS Library

An external JavaScript file - <https://personal.cs.cityu.edu.hk/~cs2204/2023/cw3cs2204.js> must be used in the **Visit Page**. The function `reserve()` in the library will be used to simulate server action. You are not allowed to use your copy in the final submission.

```
1 function reserve(date, time, no) {
2     /*
3     *data: the date of reservation
4     *time: the time of reservation
5     *no: the number of people, type: int
6     */
7     if (arguments.length != 3) {
8         alert("incorrect arguments detected"); // this is not a good way to handle error, this alert should not be activated by your code
9         return;
10    }
11    //an random int as the valid quota
12    let valid_quota = Math.floor(Math.random() * 2.0 * no);
13    if (no > valid_quota) return false // the quota is full
14    else return true; // the quota is not full
15 }
16
```

You should note that the `no` parameter should be an integer; **if something wrong happens, you should check whether you pass a non-integer or invalid value to this function.**

3. Information Page

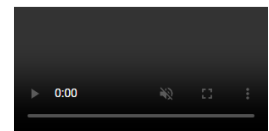
- Declare an array to hold three promotional text messages, e.g., "Optical Sensing Division has 8 openings for PhD researchers! Develop next-gen imaging systems.", and "Bio-Sensing Group offers 5 internship positions in medical sensor development for researchers!". On page load, **pick a message at random** and display it in the promotion information block. See the JavaScript Cookbook at the end of the document.

Keep every message for 3 seconds and switch to the next message. Display three messages in a loop. See the JavaScript Cookbook at the end of the document.

- Use one of the two videos below as a default in your HTML. **At the end of showing one video, switch to another and repeat this continuously.** You can use `innerHTML`.

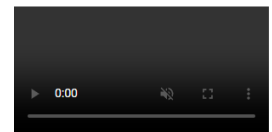
Program

Optical Sensing Division has 8 openings for PhD researchers! Develop next-gen imaging systems.



Program

Bio-Sensing Group offers 5 internship positions in medical sensor development for researchers!



4. Visit Page

- When submitting the Reservation form, check that all fields are not blank (i.e., empty string or all spaces). Display an error message "Data not completed; please re-enter" above the *middle of the box* when an error is found.
- The color of the warning message should be red.
- Do not use `alert()`; see the JavaScript cookbook.
- When **you recheck the availability**, the **previous error messages must be cleared**, and all error messages or alerts are newly displayed.

Annual Open Day Registration

Booking information:

Date: 2025/11/15
Time: 9:00-10:00am
No. of Visitors:

Data not completed, please re-enter.

Check Availability Reset

- Even if all fields are not blank, **you should also check whether the visitor number is valid, i.e., not smaller than one and is integer.** If it is not valid, you should warn the user, "Please enter a valid number of people!". Do not use *alert()*.

Annual Open Day Registration

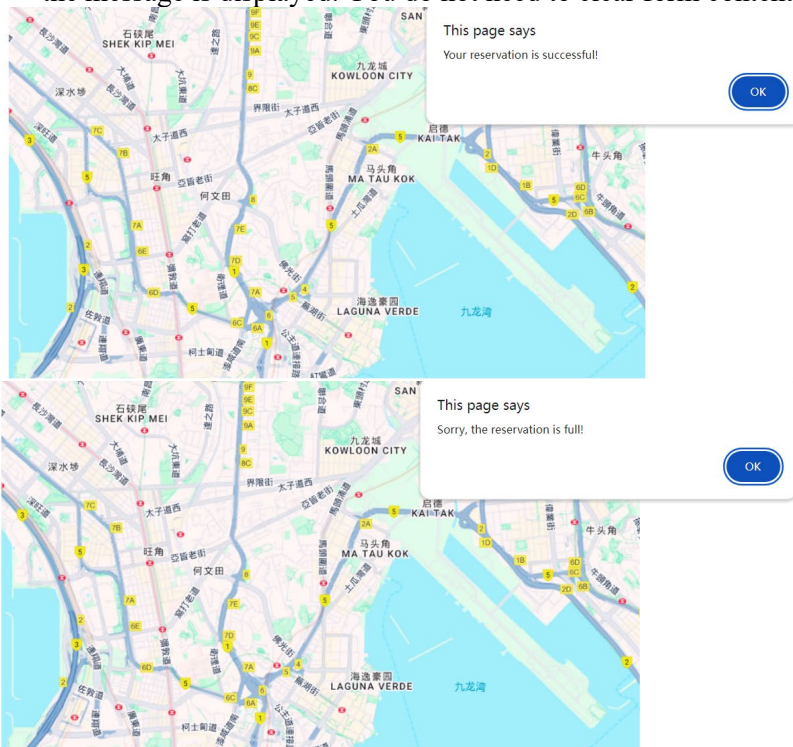
Booking information:

Date: 2025/11/15
Time: 9:00-10:00am
No. of Visitors: -1

Please enter a valid number of people!

Check Availability Reset

- If there is no error in the form data, call the function *reserve* in the library as *reserve(date, time, no-of-visitors)*. Display "Your reservation is successful!" or "Sorry, the reservation is full!" to get a function return of *true* or *false*, respectively. Cancel the form submission in all situations after the message is displayed. You do not need to clear form contents on the page.



- When you click the "Reset" button, the previous error message must be cleared.

5. Apply Page

Take away the CSS rules implementing hover and show/hide the effect of CW2 for the headings and chosen groups.

- After page loading, show the *Optical Division* heading and group application forms and set the background color of the *Optical Division* heading to white and the other two headings with another color of your choice**
- Redo show/hide effect in CW2 by using the **on-click event** for the three headings, i.e., **clicking any header will show the corresponding group application forms and hide all other group**

application forms. The background color of the clicked header will change to white, and the other two will change to your selected color.

Join our cutting-edge research teams in these focus areas:

Optical Sensing Division	Bio-Sensing Division	Smart Sensing Division
 Hyperspectral Imaging Group rank of choice	 LiDAR Development Team rank of choice	 Nanophotonics Research rank of choice

Join our cutting-edge research teams in these focus areas:

Optical Sensing Division	Bio-Sensing Division	Smart Sensing Division
 Medical Diagnostics Group rank of choice	 Environmental Sensors Team rank of choice	 Wearable Technology Lab rank of choice

Join our cutting-edge research teams in these focus areas:

Optical Sensing Division	Bio-Sensing Division	Smart Sensing Division
 IoT Sensor Networks rank of choice	 AI-Enabled Sensing rank of choice	

With JavaScript or otherwise (see JavaScript Cookbook), set up **event handlers** for all “rank of choice” buttons. You should assume the number of buttons/chosen groups is only known after the page loads. Write codes for the event handler to update the result table when clicking a button. You may modify the HTML and CSS to achieve all these things.

At first, the "Your chosen research groups" table is empty. Only the ranks show. At the beginning, in the table, there should be a rank from 1-10 in the rank column:

Your chosen research groups:

Current Internship Placements:

Division	Research Group	Rank
		1
		2
		3
		4
		5
		6
		7
		8
		9
		10

After you click the “rank of choice” button, a new choice will be added to the table according to the rank. **Note that you need special treatment for some ordinal words, such as first, second, and third. The alert message should be “You have chosen ***** as your *** chosen group in *** successfully”.**

This page says

You have chosen LiDAR Development Team as your 2nd chosen group in Optical Sensing Division successfully

OK

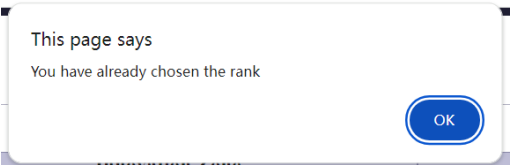
Your chosen research groups:

Current Internship Placements:

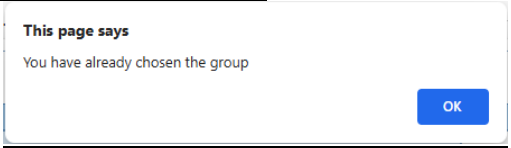
Division	Research Group	Rank
Optical Sensing Division	LiDAR Development Team	1
		2
		3
		4
		5
		6
		7
		8
		9
		10

Application Deadline: 15 October 2025

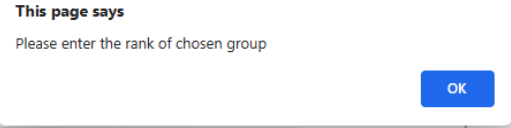
If a duplicate in rank happens, it will show, "You have already chosen this rank". Your chosen group table should reject the update:



If a duplicate in group happens, it will show, "You have already chosen this group". Your chosen group table should reject the update:

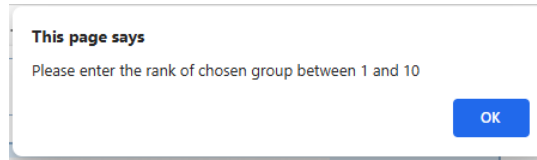


Let's be consistent: if we add a new group with duplicate groups and duplicate ranks, we only need to alert "You have already chosen this group," as shown in the screenshot above.



Also, if given an empty rank or the type of rank is not an integer, it will alert "Please enter the rank of chosen group":

If the rank of chosen is an invalid integer, which is too big or too small, you should show "Please enter the rank of chosen between 1 and 10" :



In general, the pseudo-code for checking the rank and group should be:

```
let rank_input: string = getRank(); // get rank from user input, at first the rank may be a string

// check if rank is a integer
if (isiIntegerInput(rank_input) == false) {
    alert("Please enter the rank of chosen company");
    // return;
}

let rank: number = parseInt(rank_input); // convert rank to int

// check if rank is between 1 and 10
if (rank < 1 || rank > 10) {
    alert("Please enter the rank of chosen company between 1 and 10");
    // return;
}

for(let i: number = 1; i <= 10; i++) { // go through all the chosen companies
    if (checkDuplicateCompany() == true) { // check if the company is already chosen
        alert("You have already chosen the company");
        // return;
    }
    if (checkDuplicateRank() == true) { // check if the rank is already chosen
        alert("You have already chosen the rank");
        // return;
    }
}
}
```

Hint: We can implement a JavaScript function. We first get the information about the chosen group and check the rank: check if the rank is an integer input and, if so, convert the rank to an integer from string, using the `parseInt()` function to do the conversion. If the rank is empty, null, undefined, or not an integer input, a warning alerts the user to reenter a rank. If the rank is less than 1 or greater than 10, a warning prompts the user to enter a rank between 1 and 10. Then, we check for duplicates of the group and rank of choice. Group: iterates through the numbers from 1 to 10 in a loop, checking to see if the group is the same as the one already in the list, and if there is, an alert pops up that the user has selected that group. Then, check if the user has selected the rank already. Then, determine the division affiliation, display a success message, and update the table.

- Write a function `updateTable()` to calculate the Total Number of Groups Applied and update the Last change time. The table should always be updated if you add a chosen group successfully.

Your chosen research groups:

Current Internship Placements:

Division	Research Group	Rank
Bio-Sensing Division	Medical Diagnostics Group	1
Optical Sensing Division	LiDAR Development Team	2
		3
		4
		5
		6
		7
		8
		9
		10

Total Number of Groups Applied: 2

Application Deadline: 15 October 2025

- Set up and write the `onclick` event handler for the `submit` link.

- This link is used to submit the user's chosen group. It first checks for errors; if there are any, it displays the appropriate error message.
- **If you submit an empty table, you should display the following: “You have not chosen any group.” Note that you should not use `alert()`:**

Your chosen research groups:

Current Internship Placements:

Division	Research Group	Rank
		1
		2
		3
		4
		5
		6
		7
		8
		9
		10

Total Number of Groups Applied: 0

Application Deadline: 15 October 2025

You have not chosen any groups.

- Check if there **exists a gap** in your table when you submit:
- This is a gap:

Your chosen research groups:

Current Internship Placements:

Division	Research Group	Rank
Optical Sensing Division	LiDAR Development Team	1
		2
		3
Optical Sensing Division	Nanophotonics Research	4
		5
		6
		7
		8
		9
		10

Total Number of Groups Applied: 2

Application Deadline: 15 October 2025

You have not chosen your 2nd chosen group, and 3rd chosen group, you can not leave any gap between your chosen groups.

- This is a gap:

Your chosen research groups:

Current Internship Placements:

Division	Research Group	Rank
		1
Optical Sensing Division	Hyperspectral Imaging Group	2
		3
		4
		5
		6
		7
		8
		9
		10

Total Number of Groups Applied: 1

Application Deadline: 15 October 2025

You have not chosen your 1st chosen group, you can not leave any gap between your chosen groups.

- **If you submit the table with gaps, you should display it this way: "You have not chosen your " + gap chosen groups+ ", you can not leave any gap between your chosen groups". Note that you should not use `alert()`:**
- **If the form has no problems, the successful submission message and the current time are displayed. You should display “You have successfully submitted your application at time ...”**

Your chosen research groups:

Current Internship Placements:

Division	Research Group	Rank
Optical Sensing Division	LiDAR Development Team	1
Optical Sensing Division	Hyperspectral Imaging Group	2
		3
		4
		5
		6
		7
		8
		9
		10

Total Number of Groups Applied: 2

Application Deadline: 15 October 2025

You have successfully submitted your application at time Sun Aug 24 2025 13:41:48 GMT+0800 (China Standard Time)

- Set up and write the *onclick* event handler for the *clear* link.
- **This link is used to empty the chosen groups. It empties the content for each rank. It then updates the table with the last change time and the Total Number of Groups Applied (i.e., 0).**

Your chosen research groups:

Current Internship Placements:

Division	Research Group	Rank
		1
		2
		3
		4
		5
		6
		7
		8
		9
		10

Total Number of Groups Applied: 0

Application Deadline: 15 October 2025

6. Assessment

You will be assessed by how much and how well you can apply what has been learned from the course; some considerations are:

- The requirements are met.
- Two ways of event handler setup (by JavaScript and in HTML. See the JavaScript Cookbook; must be used at least once. *State in the Design page where the two techniques are used (i.e., identify the JavaScript line no. at which the event handler is set up with what kind of method)*
- **Use of other JavaScript libraries or frameworks, such as jQuery, Angular or React.js, etc., is NOT allowed;**
- Sensible use of external, embedded, or inline scripts.
- Show a clear separation of structure, style, and script.
- Use appropriate HTML tags, classes, ids, or CSS to help your JavaScript.
- Arrange your website directories properly, e.g., HTML, style sheets, images, scripts, etc.
- CSS effect in CW2 will not be rechecked, but the error should not be seen in HTML validation.
- In June 2015, ECMAScript 6 was officially adopted as an international standard. Its goal is to make the JavaScript language usable for writing complex, large-scale applications and to become an enterprise-level development language. Currently, browser support for ES6 can be viewed at kangax.github.io/compat-table/es6/. To encourage students to engage with the latest advances in JavaScript, this assignment will give bonuses for using the following ES6 features. *Indicate on the Design page where ES6 features are used and which features are used. (i.e., indicate the JavaScript line number and explain the use of the feature to convince the Teaching Assistant that you are not just copying sample code and using it for the sake of using it).*

- (1) *let* and *const*
- (2) *set* and *map*
- (3) *class*

7. Submission

- Submit a zip file of your website with appropriate folders set up so that it can be used directly by unzipping

8. Useful techniques to complete the coursework(JavaScript Cookbook)

Multiple ways to write JavaScript for a problem may exist, with no single “correct” answer. It is essential to first think of a way to solve the problem before writing your codes.

- i) Some common techniques include but are not limited to using variables, Array/JSON to store information or DOM objects, selecting DOM objects, using loops, etc.

- ii) Get hold of elements by CSS selectors:

```
x=document.querySelector('div p')
//div p' is a string containing any valid CSS selector
//x points to the first <p> no matter how many p are selected
```

```
x=document.querySelectorAll('div p')
//x becomes an array even if one or no p is selected
```

- iii) Loop to scan through selected tags and change properties:

```
x=document.querySelectorAll('div p');
for (let i=0; i<x.length; i++) {
  x[i].style.color="red";
}
//Set the color of selected p by a loop
```

- iv) Change elements' properties, especially styles or innerHTML; use `document.write( )` method at the position where you want to create HTML in `<body>`

- v) Show or hide elements dynamically

- vi) Set up event handler

- vii) Random number generation: use methods `Math.random()` and `Math.floor()`, you can refer to https://developer.mozilla.org/en-US/docs/Web/JavaScript/Reference/Global_Objects/Math/random and https://developer.mozilla.org/en-US/docs/Web/JavaScript/Reference/Global_Objects/Math/floor .

- viii) Running JS functions continuously:

```
setInterval('myfunc()', 3000) // run myfunc() every 3 seconds
```

you can refer to <https://developer.mozilla.org/en-US/docs/Web/API/setInterval>.

- ix) The *let* declaration declares re-assignable, block-scoped local variables, initializing each to a value optionally, while *const* declares a block-scoped, read-only named constant. You can refer to https://developer.mozilla.org/en-US/docs/Web/JavaScript/Guide/Grammar_and_types.

- x) For the set, you can refer to https://developer.mozilla.org/en-US/docs/Web/JavaScript/Reference/Global_Objects/Set , and for the map, you can refer to https://developer.mozilla.org/en-US/docs/Web/JavaScript/Reference/Global_Objects/Map .
- xi) ES6 provides a closer approximation to traditional languages by introducing the concept of class. Classes can be defined using the class keyword. ES6's class can be thought of as just syntactic sugar. The new way of writing classes makes it clearer to write object prototypes in a way that is more like the syntax of object-oriented programming. You can refer to <https://developer.mozilla.org/en-US/docs/Web/JavaScript/Reference/Classes>.

```
class Person {
  constructor(name, age) {
    this.name = name;
    this.age = age;
  }
  sayHello() {
    console.log("Hello");
  }
}
class Student extends Person {
  constructor(name, age, course) {
    super(name, age);
    this.course = course;
  }
  sayHello() {
    super.sayHello();
    console.log("I am a student and I am studying " + this.course);
  }
}
```

- xii) We strongly recommend you read MDN to better understand JavaScript: <https://developer.mozilla.org/en-US/docs/Web/JavaScript>.

~ End ~